

Domain Decomposition Methods for Implicit Solvation Models

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Advanced Methods in Integral Equations and Their Applications
18th August 2026

Joint work with: Prof. Dr. Benjamin Stamm, Universität Stuttgart



1 Why Study Computational Chemistry?

2 Model Problem

3 ddPB Method

4 ddPB Derivation

5 Numerical Studies

6 Recent Work

7 Conclusions and Outlook



Why Study Computational Chemistry?

	Google Scholar	MathSciNet	
Navier Stokes Equations	874000		
Density Functional Theory			

- As of 18th August 2026



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Why Study Computational Chemistry?

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Navier Stokes Equations	874000	38364	4.3%
Density Functional Theory	2530000	5649	0.223%

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- Ionic Solvation Models¹

- Explicit Solvation Models

- Adopts molecular representation of both solute and solvent
 - Accurate results
 - Computationally expensive

¹ Zhang et. al.: *Journal of Chemical Theory and Computation*, 13 (3), 1034-1043, 2017

² Cancès, Mennucci, Tomasi: *The Journal of Chemical Physics*, 107 (8), 3032-3041, 1997

³ Honig, Nicholls: *Science*, 268 (5214), 1144-1149, 1995



- Ionic Solvation Models¹
 - Explicit Solvation Models
 - Adopts molecular representation of both solute and solvent
 - Accurate results
 - Computationally expensive
 - Implicit Solvation Models^{2,3}
 - Microscopic treatment of solute
 - Macroscopic treatment of solvent using physical properties
 - Less computational cost

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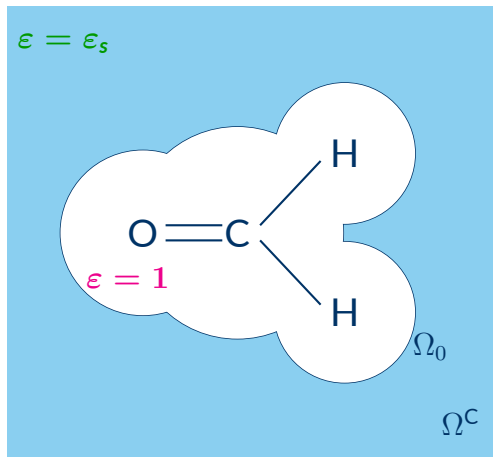


Figure 1: Formaldehyde Molecule

- Poisson-Boltzman (PB) Equation ^{1,2}

$$-\nabla \cdot \left[\varepsilon_{\text{abs}} \varepsilon(\mathbf{x}) \nabla \tilde{\psi}(\mathbf{x}) \right] = \rho^{\text{sol}}(\mathbf{x}) + \rho^{\text{ions}}(\mathbf{x}) \quad \text{in } \mathbb{R}^3$$

- $\tilde{\psi}(\mathbf{x})$: Electrostatic potential

¹ Gouy: *Journal of Physics: Theories and Applications*, 9 (1), 457-468, 1910

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- $\tilde{\psi}(\mathbf{x})$: Electrostatic potential
- $\varepsilon(\mathbf{x})$: Relative space-dependent dielectric permittivity
- $\rho^{\text{sol}}(\mathbf{x})$: Solute charge distribution

$$\rho^{\text{sol}}(\mathbf{x}) = \sum_{i=1}^M q_i \delta(\mathbf{x} - \mathbf{x}_i)$$

- M : Number of solute atoms
- q_i : Total charge on the i^{th} atom

¹ Gouy: *Journal of Physics: Theories and Applications*, 9 (1), 457-468, 1910

² Chapman: *Journal of Science*, 25 (1), 475-481, 1913



- $\rho^{\text{ions}}(\mathbf{x})$: Ionic charge distribution (Boltzmann distribution)

$$\rho^{\text{ions}}(\mathbf{x}) = \sum_{i=1}^{N_{\text{ions}}} z_i e \lambda(\mathbf{x}) c_i^{\infty} \exp \left(\frac{-z_i e \tilde{\psi}(\mathbf{x})}{K_B T} \right)$$

¹ Stein, Herbert, Head-Gordon: *The Journal of Chemical Physics*, 151 (22), 224111, 2019



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- For 1 : 1 ionic solution¹

$$\rho^{\text{ions}}(\mathbf{x}) = -2ce\lambda(\mathbf{x}) \sinh\left(\frac{e\tilde{\psi}(\mathbf{x})}{K_B T}\right)$$

— $\lambda(\mathbf{x})$: Ion-exclusion function

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- Dimensionless Poisson-Boltzman (PB) Equation

$$-\nabla \cdot [\epsilon(\mathbf{x}) \nabla \psi(\mathbf{x})] + \kappa^2 \epsilon_s \lambda(\mathbf{x}) \sinh(\psi(\mathbf{x})) = \frac{1}{\beta \epsilon_{\text{abs}}} \rho^{\text{sol}}(\mathbf{x}) \quad \text{in } \mathbb{R}^3$$

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- $\psi(\mathbf{x}) : \tilde{\psi}(\mathbf{x})\beta$
- κ : Debye Hückel Screening Constant¹
- $\beta : e/K_B T$

¹Debye, Hückel: *Physikalische Zeitschrift*, 24 (9), 185-206, 1923

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- For a strongly **charged** molecule $|\psi(\mathbf{x})| \not\ll 1$



Solute Cavity

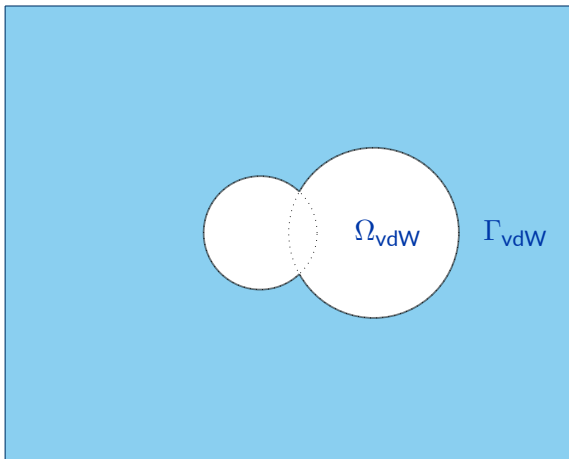


Figure 2: Solute probes and solute-solvent boundary for a diatomic molecule.

- **vdW:** van-der Waal's Surface

Solute Cavity

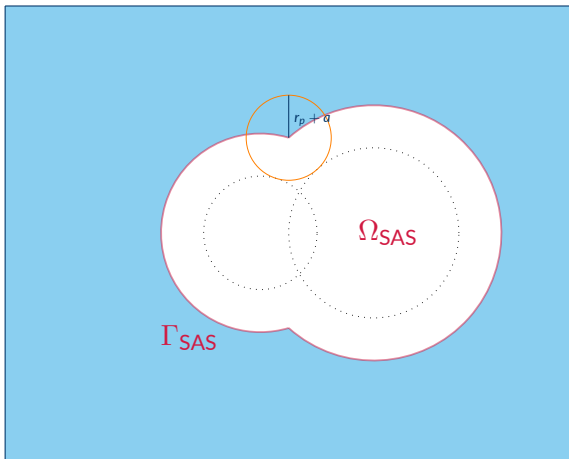


Figure 2: Solute probes and solute-solvent boundary for a diatomic molecule.

- **SAS:** Solvent Accessible Surface

Solute Cavity

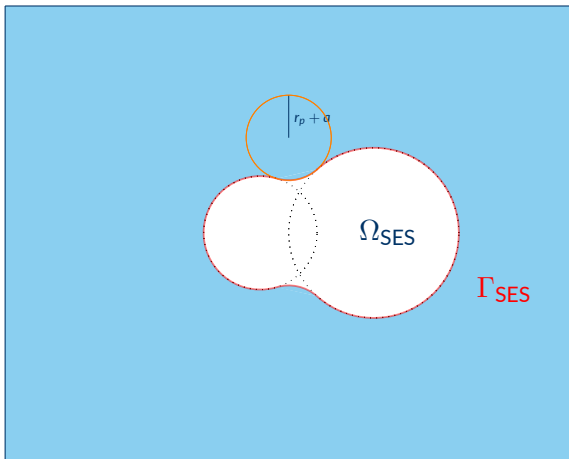


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- **SES:** Solvent Excluded Surface

Solute Cavity

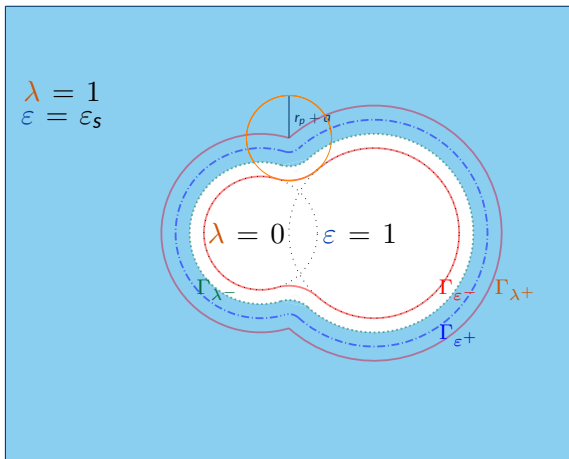


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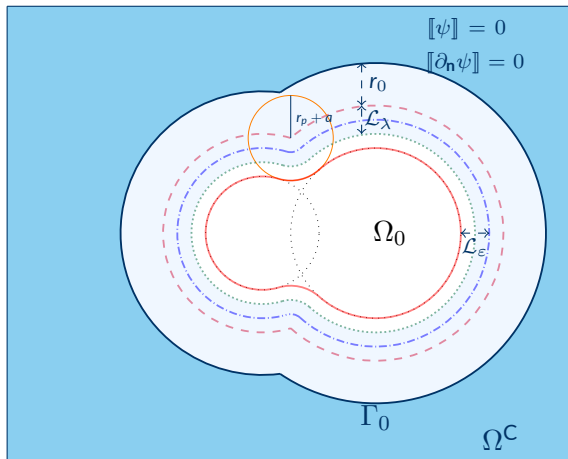


Figure 2: Solute probes and solute-solvent boundary for a diatomic molecule.

- Dielectric Permittivity Function¹

$$\varepsilon(\mathbf{x}) = \begin{cases} 1 & \mathbf{x} \in \Omega_{\text{SES}}, \\ 1 + (\varepsilon_s - 1)\xi \left(\frac{f_{\text{SAS}}(\mathbf{x}) + r_p + a}{r_p} \right) & \mathbf{x} \in \mathcal{L}_\varepsilon, \\ \varepsilon_s & \text{else,} \end{cases}$$

¹ Quan, Stamm: *Journal of Computational Physics*, 322 (1), 760-782, 2016

² Stern: *Zeitschrift für Elektrochemie*, 30 (508), 21-22, 1924



Permittivity and Ion-Exclusion Function

- Dielectric Permittivity Function¹

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- Ion-Exclusion Function²

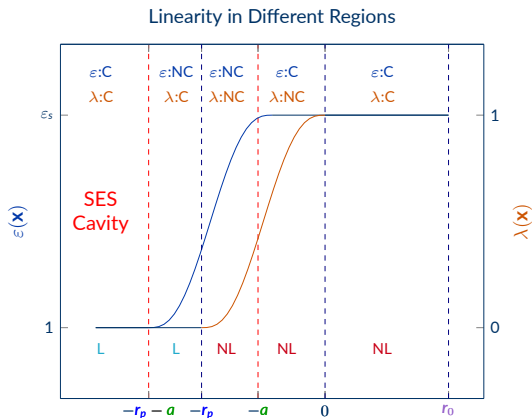
$$\lambda(\mathbf{x}) = \begin{cases} 0 & \mathbf{x} \in \Omega_{\lambda-}, \\ \xi \left(\frac{f_{\text{SAS}}(\mathbf{x}) + r_p}{r_p} \right) & \mathbf{x} \in \mathcal{L}_\lambda, \\ 1 & \text{else,} \end{cases}$$

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Permittivity and Ion-Exclusion Function



- C: Constant, NC: Non-Constant
- L: Linear, NL: Non-Linear



- Why the Problem is Numerically Awkward?
 - Unbounded Exterior Domain
 - Spatially Varying Coefficients
 - Nonlinear Ionic Term
 - Complicated SES Geometry



- Why the Problem is Numerically Awkward?
 - Unbounded Exterior Domain
 - Spatially Varying Coefficients
 - Nonlinear Ionic Term
 - Complicated SES Geometry
- Key Idea: Replace unbounded problem with a coupled problem



- The PB equation can be written in two equations

$$\begin{aligned} -\nabla \cdot [\varepsilon(\mathbf{x}) \nabla \psi(\mathbf{x})] + \kappa^2 \varepsilon_s \lambda(\mathbf{x}) \sinh(\psi(\mathbf{x})) &= \frac{1}{\beta \varepsilon_{\text{abs}}} \rho^{\text{sol}}(\mathbf{x}) && \text{in } \Omega_0, \\ -\Delta \psi(\mathbf{x}) + \kappa^2 \psi(\mathbf{x}) &= 0 && \text{in } \Omega^c, \end{aligned}$$

with

$$\begin{aligned} \llbracket \psi \rrbracket &= 0, \\ \llbracket \partial_{\mathbf{n}} \psi \rrbracket &= 0 \quad \text{on } \Gamma_0 := \partial\Omega_0, \end{aligned}$$



- Using **Potential Theory** the final equations are

$$\begin{aligned} -\nabla \cdot [\varepsilon(\mathbf{x}) \nabla \psi_r(\mathbf{x})] + \kappa^2 \varepsilon_s \lambda(\mathbf{x}) \mathcal{F}(\psi_r + \psi_0)(\psi_r + \psi_0)(\mathbf{x}) \\ = \nabla \cdot [(\varepsilon(\mathbf{x}) - 1) \nabla \psi_0(\mathbf{x})] \quad \text{in } \Omega_0 \quad [\text{GSP}] \end{aligned}$$



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$$\begin{aligned} \psi_0 + \psi_r &= \psi_e \quad \text{on } \Gamma_0, \\ \psi_e &= S_{\kappa} \sigma_e \quad \text{on } \Gamma_0 \end{aligned}$$



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where

- ψ_r : Reaction potential in Ω
- ψ_0 : Potential generated by ρ_M satisfying,

$$-\Delta \psi_0 = \frac{1}{\beta \varepsilon_{\text{abs}}} \rho_M$$



- ψ_e : Extended potential from Ω^C to Ω_0

¹Sauter, Schwab, Springer, Berlin-2011, 101-181



- ψ_e : Extended potential from Ω^C to Ω_0
- $\mathcal{F}(\Phi) = \frac{\sinh(\Phi)}{\Phi}$

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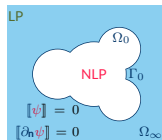
- ψ_e : Extended potential from Ω^C to Ω_0
- $\mathcal{F}(\Phi) = \frac{\sinh(\Phi)}{\Phi}$
- σ_e : Charge density generating ψ_e satisfying

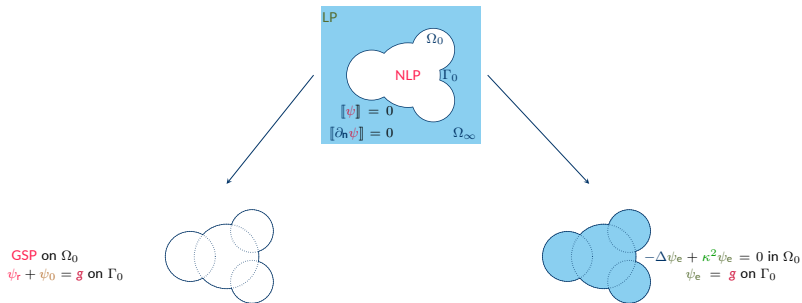
$$S_{\kappa} \sigma_e(\mathbf{x}) = \int_{\Gamma_0} \frac{\exp(-\kappa|\mathbf{x} - \mathbf{y}|) \sigma_e(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|} = \psi_e \quad \forall \mathbf{x} \in \Gamma_0$$

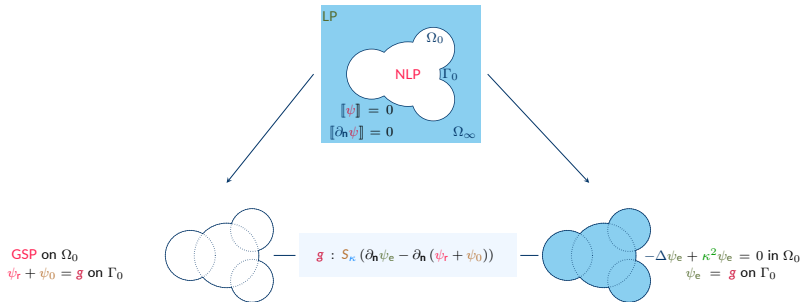
- S_{κ} : Invertible single-layer potential operator ¹

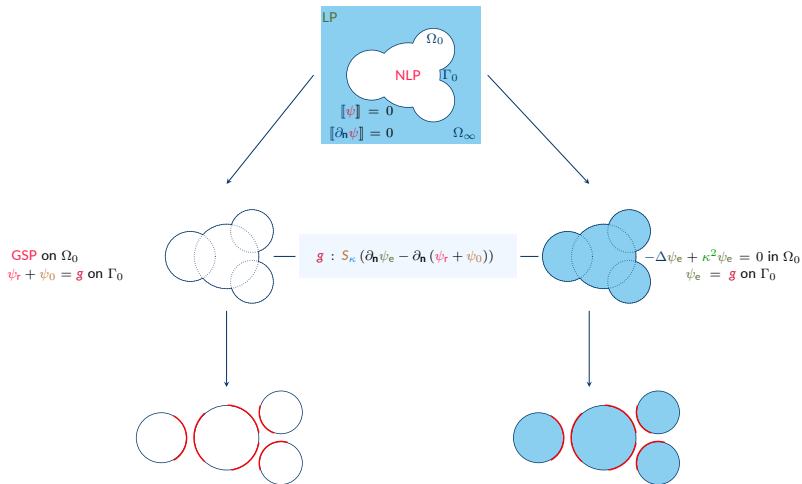
$$S_{\kappa} : H^{-1/2}(\Gamma_0) \rightarrow H^{1/2}(\Gamma_0)$$

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- We can decompose Ω_0

$$\Omega_0 = \bigcup_{j=1}^M \Omega_j, \quad \Omega_j = B_{R_j}(\mathbf{x}_j)$$

- $R_j = r_j + a + r_0 + r_p$

¹ Cances, Maday, Stamm: *The Journal of Chemical Physics*, 139 (5), 054111, 2013

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⁴ Stamm, Cances, Lipparini, Maday: *The Journal of Chemical Physics*, 144 (1), 054101, 2016

⁵ Gatto, Lipparini, Stamm: *The Journal of Chemical Physics*, 147 (1), 224108, 2017

⁶ Quan, Stamm, Maday: *SIAM Journal on Scientific Computing*, 41 (2), B320-B350, 2019

⁷ J., Nottoli, Mikhalev, Quan, Stamm: *The Journal of Chemical Physics*, 158 (1), 104105, 2023

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- GSP equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\epsilon}(\mathbf{x}) \nabla u(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{u}(\mathbf{x})) u(\mathbf{x}) &= f(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ u(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$



Single Domain Solvers

- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\epsilon}(\mathbf{x}) \nabla \mathbf{u}(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x})) \mathbf{u}(\mathbf{x}) &= \mathbf{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ \mathbf{u}(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- **Transformation** to Homogeneous Problem

$$\begin{aligned} -\nabla \cdot [\tilde{\epsilon}(\mathbf{x}) \nabla \mathbf{w}(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{\mathbf{w} + \hat{\mathbf{u}}_1}\right)(\mathbf{x})\right) \mathbf{w}(\mathbf{x}) &= \tilde{\mathbf{f}}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ \mathbf{w}(\mathbf{x}) &= 0 \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$



- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla \mathbf{u}(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{\mathbf{u}}(\mathbf{x})) \mathbf{u}(\mathbf{x}) &= \mathbf{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ \mathbf{u}(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

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- $\mathbf{w}(\mathbf{x}) = \mathbf{u}(\mathbf{x}) - \hat{\mathbf{u}}_1(\mathbf{x})$
- $\tilde{\mathbf{f}}(\mathbf{x}) = \mathbf{f}(\mathbf{x}) + \nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla \hat{\mathbf{u}}_1(\mathbf{x})] - \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{\mathbf{w} + \hat{\mathbf{u}}_1}\right)(\mathbf{x})\right) \hat{\mathbf{u}}_1(\mathbf{x})$



- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla u(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{u}(\mathbf{x})) u(\mathbf{x}) &= f(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ u(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- **Transformation** to Homogeneous Problem

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{w + \hat{u}_1}\right)(\mathbf{x})\right) w(\mathbf{x}) &= \tilde{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ w(\mathbf{x}) &= 0 \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- $w(\mathbf{x}) = u(\mathbf{x}) - \hat{u}_1(\mathbf{x})$
- $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + \nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla \hat{u}_1(\mathbf{x})] - \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{w + \hat{u}_1}\right)(\mathbf{x})\right) \hat{u}_1(\mathbf{x})$
- $\hat{u}_1(\mathbf{x})$: Laplace solution satisfying the boundary condition



- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla u(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{u}(\mathbf{x})) u(\mathbf{x}) &= f(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ u(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- **Transformation** to Homogeneous Problem

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{w + \hat{u}_1}\right)(\mathbf{x})\right) w(\mathbf{x}) &= \tilde{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ w(\mathbf{x}) &= 0 \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

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 - $\hat{u}_1(\mathbf{x})$: Laplace solution satisfying the boundary condition
- $B_{r_j}(\mathbf{x}_j) \subset \Omega_j$



- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla u(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{u}(\mathbf{x})) u(\mathbf{x}) &= f(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ u(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- **Transformation** to Homogeneous Problem

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\overline{(w + \hat{u}_1)}(\mathbf{x})\right) w(\mathbf{x}) &= \tilde{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ w(\mathbf{x}) &= 0 \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- $w(\mathbf{x}) = u(\mathbf{x}) - \hat{u}_1(\mathbf{x})$
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 - $\hat{u}_1(\mathbf{x})$: Laplace solution satisfying the boundary condition
- $B_{r_j}(\mathbf{x}_j) \subset \Omega_j$
 - $\psi_r(\mathbf{x})$ is harmonic in $B_{r_j}(\mathbf{x}_j)$



Single Domain Solvers

- **GSP** equation in unit ball

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla u(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{u}(\mathbf{x})) u(\mathbf{x}) &= f(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ u(\mathbf{x}) &= \phi_r(\mathbf{x}) \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- **Transformation** to Homogeneous Problem

$$\begin{aligned} -\nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x})] + \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{w + \hat{u}_1}\right)(\mathbf{x})\right) w(\mathbf{x}) &= \tilde{f}(\mathbf{x}) \quad \text{in } B_1(\mathbf{0}), \\ w(\mathbf{x}) &= 0 \quad \text{on } \partial B_1(\mathbf{0}) \end{aligned}$$

- $w(\mathbf{x}) = u(\mathbf{x}) - \hat{u}_1(\mathbf{x})$
 - $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + \nabla \cdot [\tilde{\varepsilon}(\mathbf{x}) \nabla \hat{u}_1(\mathbf{x})] - \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}\left(\left(\overline{w + \hat{u}_1}\right)(\mathbf{x})\right) \hat{u}_1(\mathbf{x})$
 - $\hat{u}_1(\mathbf{x})$: Laplace solution satisfying the boundary condition
- $B_{r_j}(\mathbf{x}_j) \subset \Omega_j$
 - $\psi_r(\mathbf{x})$ is harmonic in $B_{r_j}(\mathbf{x}_j)$
 - $w(\mathbf{x})$ is harmonic in $B_\delta(\mathbf{0})$ where

$$\delta = \frac{r_j}{r_j + r_0 + r_p + a} \in (0, 1)$$



- Find $w \in H_{0,\delta}^1(\mathcal{D})$ such that

$$\begin{aligned} \int_{\mathcal{D}} \tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x}) \nabla \tilde{w}(\mathbf{x}) + \int_{\mathcal{D}} \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{w}(\mathbf{x})) w(\mathbf{x}) \tilde{w}(\mathbf{x}) \\ + \int_{\partial B_\delta(\mathbf{0})} (\mathcal{T}w) \tilde{w}(\mathbf{x}) = \int_{\mathcal{D}} \tilde{f}(\mathbf{x}) \tilde{w}(\mathbf{x}) \quad \forall \tilde{w} \in H_{0,\delta}^1(\mathcal{D}), \end{aligned}$$



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- $\mathcal{D} = B_1(\mathbf{0}) \setminus B_\delta(\mathbf{0})$
- $H_{0,\delta}^1(\mathcal{D}) = \{w \in H^1(\mathcal{D}) : w|_{\partial B_1(\mathbf{0})} = 0\}$



- Find $w \in H_{0,\delta}^1(\mathcal{D})$ such that

$$\int_{\mathcal{D}} \tilde{\varepsilon}(\mathbf{x}) \nabla w(\mathbf{x}) \nabla \tilde{w}(\mathbf{x}) + \int_{\mathcal{D}} \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{w}(\mathbf{x})) w(\mathbf{x}) \tilde{w}(\mathbf{x}) + \int_{\partial B_\delta(\mathbf{0})} (\mathcal{T}w) \tilde{w}(\mathbf{x}) = \int_{\mathcal{D}} \tilde{f}(\mathbf{x}) \tilde{w}(\mathbf{x}) \quad \forall \tilde{w} \in H_{0,\delta}^1(\mathcal{D}),$$

- $\mathcal{D} = B_1(\mathbf{0}) \setminus B_\delta(\mathbf{0})$
 - $H_{0,\delta}^1(\mathcal{D}) = \{w \in H^1(\mathcal{D}) : w|_{\partial B_1(\mathbf{0})} = 0\}$
- Using Galerkin discretisation

$$w_B(r, \theta, \varphi) = \sum_{i=0}^N \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} [\phi_r]_{i\ell}^m q_i(r) Y_\ell^m(\theta, \varphi) \quad \forall \delta \leq r \leq 1; \quad 0 \leq \theta \leq \pi; \quad 0 \leq \varphi \leq 2\pi,$$

- q_i : Legendre polynomial of order i
- N : Maximum degree of Legendre polynomial of order q_i
- Y_ℓ^m : Spherical Harmonic Basis
- $\ell_{\max}$: Maximum degree of Y_ℓ^m



- System of Equation

$$\mathbf{A}\mathbf{X}_r = \mathbf{F}$$

where

- $k := N(\ell^2 + m + 1) + i \in \{1, 2, \dots, N(\ell_{\max} + 1)^2\}$, k' entry

$$\begin{aligned} [\mathbf{A}]_{k,k'} &= \int_{\mathcal{D}} \tilde{\varepsilon}(\mathbf{x}) \nabla (\varrho_i \mathbf{Y}_{\ell}^m) \cdot \nabla (\varrho_j \mathbf{Y}_{\ell'}^{m'}) \\ &\quad + \int_{\mathcal{D}} \tilde{\lambda}(\mathbf{x}) \tilde{\mathcal{F}}(\bar{\mathbf{w}}(\mathbf{x})) \varrho_i \mathbf{Y}_{\ell}^m \varrho_j \mathbf{Y}_{\ell'}^{m'} \\ &\quad + \frac{\ell}{\delta} \int_{\partial B_{\delta}(\mathbf{0})} \varrho_i \mathbf{Y}_{\ell}^m \varrho_j \mathbf{Y}_{\ell'}^{m'}, \end{aligned}$$

○

$$[\mathbf{F}]_k = \int_{\mathcal{D}} \tilde{f} \varrho_j \mathbf{Y}_{\ell'}^{m'} \quad \forall k \in \{1, \dots, N(\ell_{\max} + 1)^2\}.$$



- HSP equation in unit ball ¹

$$\begin{aligned} -\Delta u_e + \kappa^2 u_e^2 &= 0 && \text{in } B_1(0), \\ u_e &= \phi_e && \text{on } \mathbb{S}^2 \end{aligned}$$

- u_e can be numerically approximated by $\tilde{u}_e$

$$\tilde{u}_e(r, \theta, \varphi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} \left[\tilde{\phi}_e \right]_{\ell}^m \frac{i_{\ell}(r)}{i_{\ell}(1)} Y_{\ell}^m(\theta, \varphi)$$

for $0 \leq r \leq 1$, $0 \leq \theta \leq \pi$, $0 \leq \varphi < 2\pi$

¹Quan, Stamm, Maday: *SIAM Journal on Scientific Computing*, 41 (2), B320-B350, 2019

²Lebedev: *USSR Computational Mathematics and Mathematical Physics*, 16 (2), 293-306, 1976



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for $0 \leq r \leq 1$, $0 \leq \theta \leq \pi$, $0 \leq \varphi < 2\pi$

- $\left[\tilde{\phi}_e \right]_{\ell}^m$: Numerical approximation of $\left[\phi_e \right]_{\ell}^{m2}$

$$\left[\tilde{\phi}_e \right]_{\ell}^m = \sum_{n=1}^{N_{\text{leb}}} \omega_n^{\text{leb}} \phi_e(s_n) Y_{\ell}^m(s_n)$$

¹Quan, Stamm, Maday: *SIAM Journal on Scientific Computing*, 41 (2), B320-B350, 2019

²Lebedev: *USSR Computational Mathematics and Mathematical Physics*, 16 (2), 293-306, 1976



- Numerical Integration^{1,2}

¹ Haxton: *J.Phys.B*, 40 (1), 4443, 2007

² Parter: *Journal of Scientific Computing*, 14 (1), 347-355, 1999



- Numerical Integration^{1,2}

$$\begin{aligned} \int_{\mathcal{D}} h(\mathbf{x}) d\mathbf{x} &= \int_{\delta}^1 r^2 \int_{\mathbb{S}^2} h(r, \mathbf{s}) d\mathbf{s} dr \\ &\approx \frac{1-\delta}{2} \sum_{m=1}^{N_{\text{lgI}}} \sum_{n=1}^{N_{\text{leb}}} \omega_m^{\text{lgI}} \omega_n^{\text{leb}} \left(\frac{1-\delta}{2} (x_m + 1) + \delta \right)^2 \\ &\quad \times h \left(\frac{1-\delta}{2} (x_m + 1) + \delta, \mathbf{s}_n \right). \end{aligned}$$

¹ Haxton: *J.Phys.B*, 40 (1), 4443, 2007

² Parter: *Journal of Scientific Computing*, 14 (1), 347-355, 1999



- Energy Computation¹

$$E_s = \frac{\beta}{2} \int_{\Omega} \rho^{\text{sol}}(\mathbf{x}) \psi_r(\mathbf{x}) + \frac{\beta^2 \kappa^2 \varepsilon_s}{8\pi} \int_{\Omega} \lambda(\mathbf{x}) (\psi_r(\mathbf{x}) \sinh(\psi_r(\mathbf{x})) - 2 \cosh(\psi_r(\mathbf{x})))$$

¹Stein, Herbert, Head-Gordon: *The Journal of Chemical Physics*, 151 (22), 224111, 2019



- Energy Computation¹

$$E_s = \frac{\beta}{2} \int_{\Omega} \rho^{\text{sol}}(\mathbf{x}) \psi_r(\mathbf{x}) + \frac{\beta^2 \kappa^2 \varepsilon_s}{8\pi} \int_{\Omega} \lambda(\mathbf{x}) (\psi_r(\mathbf{x}) \sinh(\psi_r(\mathbf{x})) - 2 \cosh(\psi_r(\mathbf{x})))$$

- Stopping Criteria

- Global Iterative Process

$$|E_s^k - E_s^{k-1}| / |E_s^k| \leq \text{tol}$$

¹Stein, Herbert, Head-Gordon: *The Journal of Chemical Physics*, 151 (22), 224111, 2019



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- Stopping Criteria

 - Global Iterative Process

$$|E_s^k - E_s^{k-1}| / |E_s^k| \leq \text{tol}$$

 - DD loop

$$\frac{\|X_r^k - X_r^{k-1}\|_{\ell^2}}{\|X_r^k\|_{\ell^2}} \leq 10 \times \text{tol}$$

¹Stein, Herbert, Head-Gordon: *The Journal of Chemical Physics*, 151 (22), 224111, 2019

- Energy Computation¹

$$E_s = \frac{\beta}{2} \int_{\Omega} \rho^{\text{sol}}(\mathbf{x}) \psi_r(\mathbf{x}) + \frac{\beta^2 \kappa^2 \varepsilon_s}{8\pi} \int_{\Omega} \lambda(\mathbf{x}) (\psi_r(\mathbf{x}) \sinh(\psi_r(\mathbf{x})) - 2 \cosh(\psi_r(\mathbf{x})))$$

- Stopping Criteria

 - Global Iterative Process

$$|E_s^k - E_s^{k-1}| / |E_s^k| \leq \text{tol}$$

 - DD loop

$$\frac{\|X_r^k - X_r^{k-1}\|_{\ell^2}}{\|X_r^k\|_{\ell^2}} \leq 10 \times \text{tol}$$

 - Matrix loop

$$\frac{\|X_{r,i}^k - X_{r,i}^{k-1}\|_{\ell^2}}{\|X_{r,i}^k\|_{\ell^2}} \leq 100 \times \text{tol}$$

¹Stein, Herbert, Head-Gordon: *The Journal of Chemical Physics*, 151 (22), 224111, 2019



- Constants in the model



- Constants in the model

- ϵ_s : 78.54
- κ : $0.104 \text{ \AA}^{-1}$
- r_p : $1.4 \text{ \AA}$
- T : 298.15 K
- tol : 10^{-7}

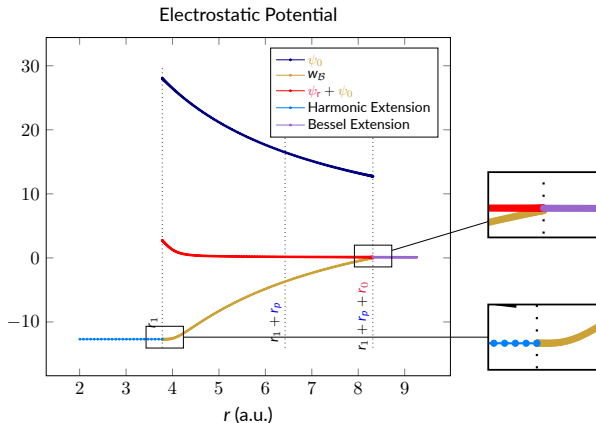


- **Constants** in the model
 - ϵ_s : 78.54
 - κ : $0.104 \text{ \AA}^{-1}$
 - r_p : $1.4 \text{ \AA}$
 - T : 298.15 K
 - tol : 10^{-7}
 - **Conversion** to atomic units



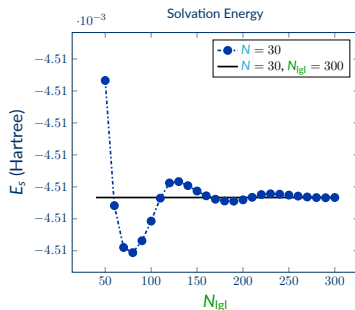
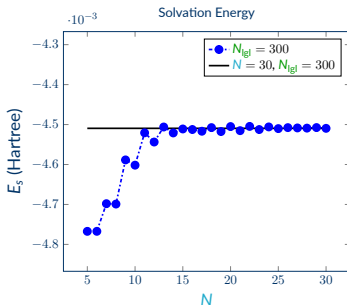
Potential for One Sphere

- One Sphere
- Discretisation Parameters: $N = 20$, $N_{|g|} = 200$
- Geometric Parameters: $r_1 = 2 \text{ \AA}$, $r_0 = 1 \text{ \AA}$, $a = 0 \text{ \AA}$



Effect of Discretisation Parameters

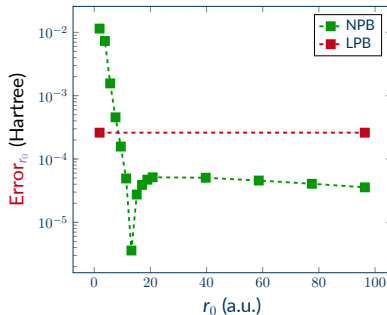
- Discretisation Parameters: $N = 30$, $N_{\text{lg}} = 300$
- Geometric Parameters: $r_1 = 2 \text{ \AA}$, $r_0 = 5 \text{ \AA}$, $a = 0 \text{ \AA}$



- Similar observations for spherical discretisation



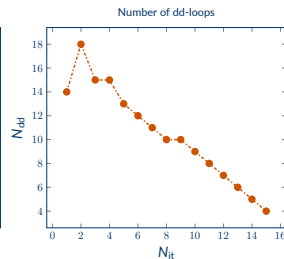
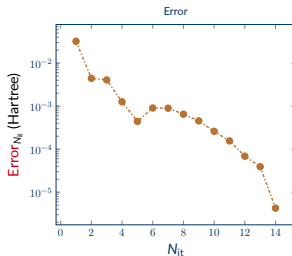
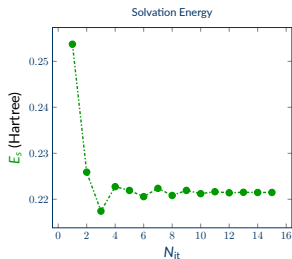
- Hydrogen Fluoride Molecule
- Discretisation Parameters: $N = 40$, $N_{\text{lg}} = 500$, $\ell_{\text{max}} = 7$, $N_{\text{leb}} = 86$
- Geometric Parameters: $a = 1 \text{ \AA}$, $r_0^{\text{max}} = 91 \text{ \AA}$



- $\text{Error}_{r_0} := |E_s(r_0) - E_s(r_0^{\text{max}})|$

Convergence of Global Strategy

- Caffeine Molecule
- Discretisation Parameters: $N = 15$, $N_{\text{gl}} = 50$, $\ell_{\text{max}} = 9$, $N_{\text{leb}} = 350$
- Geometric Parameters: $r_0 = 5 \text{ \AA}$, $a = 1 \text{ \AA}$

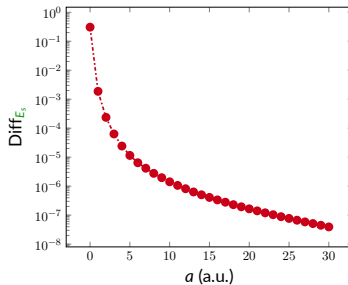


- $\text{Error}_{N_{\text{it}}} := |E_s^\infty - E_s^{N_{\text{it}}}|$
 - E_s^∞ : 15 Outer Iterations



Effect of Stern layer

- Hydrogen Fluoride Molecule
- Discretisation Parameters: $N = 15$, $N_{|g|} = 50$, $\ell_{\max} = 9$, $N_{\text{leb}} = 350$
- Geometric Parameters: $r_0 = 2 \text{ \AA}$

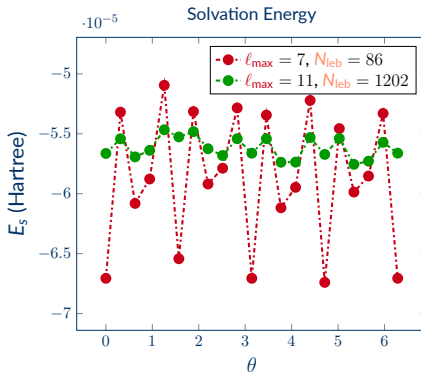


- $$\text{Diff}_{E_s} = \left| \frac{\beta^2 \kappa^2 \epsilon_s}{8\pi} \int_{\Omega_0} \lambda(\mathbf{x}) \left[\psi_r(\mathbf{x}) \sinh(\psi_r(\mathbf{x})) - 2 \{ \cosh(\psi_r(\mathbf{x})) - 1 \} \right] \right|,$$



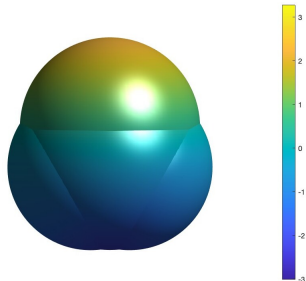
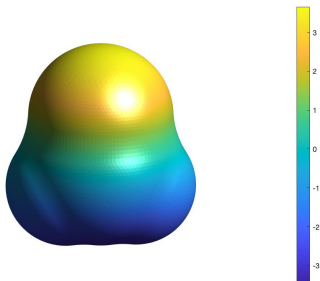
Rotational Symmetry

- Hydrogen Fluoride Molecule
- Discretisation Parameters: $N = 15$, $N_{|g|} = 50$
- Geometric Parameters: $r_0 = 3 \text{ \AA}$, $a = 5 \text{ \AA}$

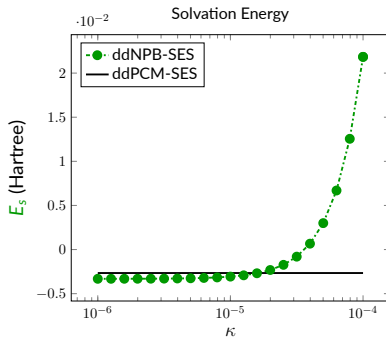


Visualisation of Potential

- Visualisation of ψ_r
- Discretisation Parameters: $N = 15$, $N_{|g|} = 50$, $\ell_{\max} = 11$,
 $N_{\text{leb}} = 1202$
- Geometric Parameters: $r_0 = 3 \text{ \AA}$, $a = 1 \text{ \AA}$



- Hydrogen Fluoride Molecule
- Variation of κ
- Discretisation Parameters: $N = 15$, $N_{\text{lg}} = 30$, $\ell_{\text{max}} = 7$, $N_{\text{leb}} = 86$
- Geometric Parameters: $r_0 = 0 \text{ \AA}$, $a = 0 \text{ \AA}$



- COSMO Equations

$$\begin{aligned} -\Delta\psi(\mathbf{x}) &= 0 && \text{in } \Omega_0, \\ \psi(\mathbf{x}) &= \psi_0(\mathbf{x}) && \text{on } \Gamma_0 \end{aligned}$$

- COSMO Equations

$$\begin{aligned} -\Delta\psi(\mathbf{x}) &= 0 && \text{in } \Omega_0, \\ \psi(\mathbf{x}) &= \psi_0(\mathbf{x}) && \text{on } \Gamma_0 \end{aligned}$$

- Well-posedness of the ddCOSMO for L^2 boundary data
- Explicit Geometry dependent Convergence Rates
- Error Estimates
- Sharp Convergence rates for two-disk



- Key Analytical Ideas¹

- Dirichlet-to-Dirichlet Map(DtD)

- Boundary Data on One Disk
 - Harmonic Extension
 - Trace on Interior Circular Arc

¹Corso, Hassan, J., Stamm: *SIAM Journal on Mathematical Analysis*, 58 (2), 1571-1612, 2026

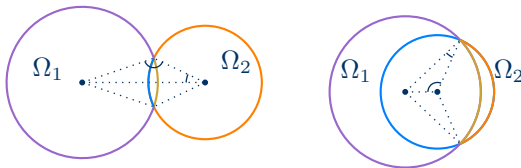


- Key Analytical Ideas¹
 - Dirichlet-to-Dirichlet Map(DtD)
 - Boundary Data on One Disk
 - Harmonic Extension
 - Trace on Interior Circular Arc
 - Analysis Characterises the DtD Map
 - Numerical Radius bounds to control Parallel Schwarz

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- Key Analytical Ideas¹
 - Dirichlet-to-Dirichlet Map(DtD)
 - Boundary Data on One Disk
 - Harmonic Extension
 - Trace on Interior Circular Arc
 - Analysis Characterises the DtD Map
 - Numerical Radius bounds to control Parallel Schwarz
- Convergence Rates **deteriorates** as the overlap becomes less effective



¹Corso, Hassan, J., Stamm: *SIAM Journal on Mathematical Analysis*, 58 (2), 1571-1612, 2026

- Conclusions¹
 - Formulation of domain decomposition method for PB equations

¹ J., Stamm: arXiv:2309.06862, 2025

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Thank You!

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Webpage: tinyurl.com/abhi0207



¹ JRF Position to be announced soon

